Name: _____ Date: _____ Class: _____

MyDesign Portfolio

Element D Initial Template

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a decision matrix.

Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document.

Generate a design blueprint with enough details that someone else not familiar with your project could also build your design. Think of this as your first sketch.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

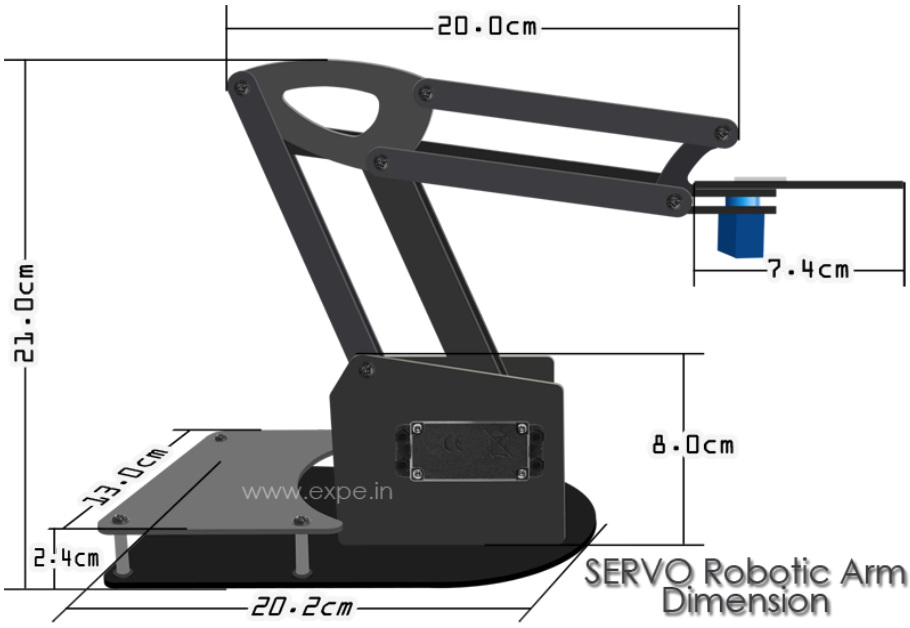
This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Name: _____ **Date:** _____ **Class:** _____

Test Tube Robotic Close

Date: 3/4/25
Description: It will be made out of metal and the size of a small test tube. The robotic arms will be welded on and made out of metal.

https://www.tubixtestube.com/37mm-flat-bottom-glass-test-tube-with-aluminum-cap



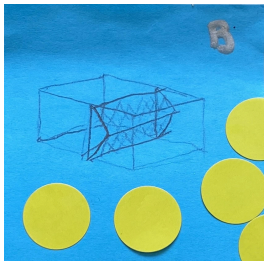
https://www.google.com/url?sa=i&url=https%3A%2F%2Fexpe.in%2Fservo-robotic-arm%2F&psig=AOvVaw0Kw5lhpo9PkiyU0LpT_ulS&ust=1741201509147000&source=images&cd=vfe&opi=89978449&ved=0CBQQjhxqFwoTCIDXy56P8YsDFQAAAAAdAAAAABAE

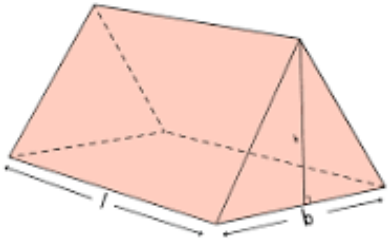
Horizontal Bulge Double Filter

Name: _____ **Date:** _____ **Class:** _____

Date: 3/4/2025
Description: A double filter, just larger than and just smaller than microplastics, set up to cover the front face of the ROV with a horizontal bulge to deflect anything too large to pass through the filter. Sides need to be sealed to prevent microplastic samples from being lost. Also to collect the samples we need easy access to the middle of both filters.

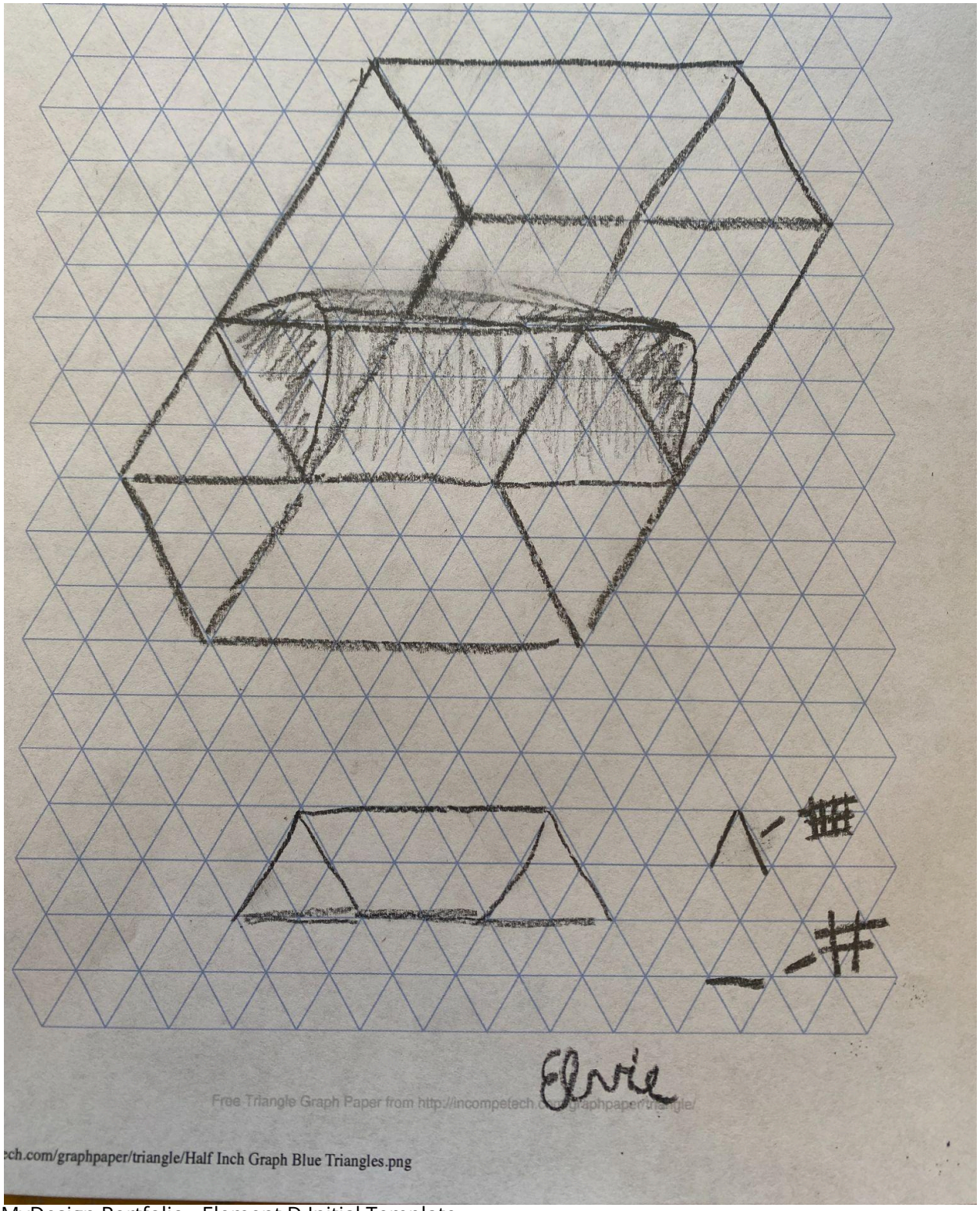
Drawing that includes distinguishing information such as scale, dimensions, and materials:



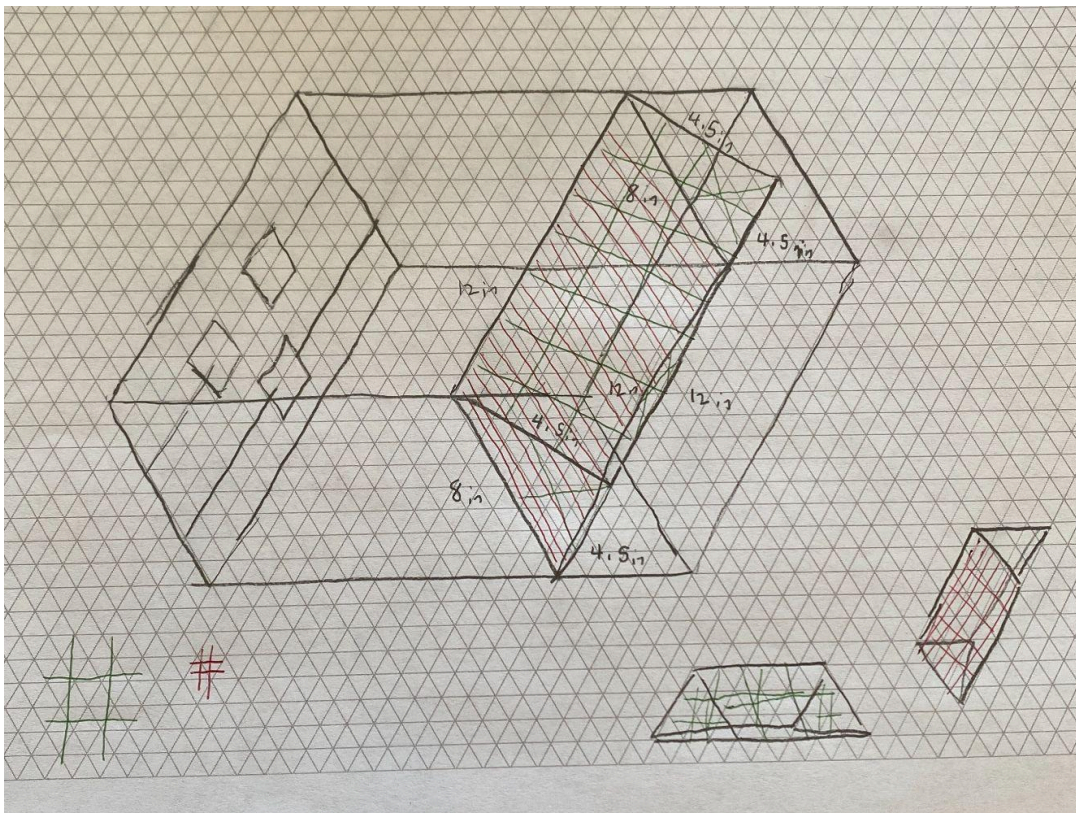


https://www.geeksforgeeks.org/triangular-prism/

Name: _____ **Date:** _____ **Class:** _____



Name: _____ **Date:** _____ **Class:** _____



Design Idea Comparison and Selection *(D1, D2)*

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify lothe weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

Name: _____ **Date:** _____ **Class:** _____

Criteria	weight	Net	Test Tube
Sample Sizes	0.4	1.0	0.3
Integration *	0.15	0.6	0.45
Reusability	0.2	0.4	0.7
Ease of Collection	0.15	0.4	0.7
Affordability	0.07	0.55	0.3
Uniqueness	0.03	0.3	0.4
score		0.68	0.47

* how well it attaches to ROV

Design Blueprint *(D4)*

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date: 3/6/25
Sketch:
Manufacturing Methods:
    Step 1:
    Step 2:
    Step 3:
Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
Net (Large and small filter)		1 sq ft per net	
3D printer filament		4 ½ in brackets worth	
PVC Pipe		4 feet (estimate)	

Name: _____ **Date:** _____ **Class:** _____

Total Cost			

Plan of Action *(D3)*

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

- Frame modifications (Happens in parallel with net preparation, so due ≈3/24 but can likely be finished before the net arives, all dependant on the materials though.)
    -
    - 3D print ½ in PVC pipe brackets
- Cutting the net (Due ≈3/24 depending on how quickly the parts ship)
    - 2 Front panels
    - 1 Back panel
- Attaching the net (Due 3/28)
    - Removable back